3 Yr. Degree/4 Yr. Honours 4th Semester Examination, 2025 (CCFUP)

Subject : Mathematics

Course : MATH4021 (MINOR)

(Ordinary Differential Equations)

Time: 3 Hours **Full Marks: 60**

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Notation and symbols have their usual meaning.

1. Answer *any ten* questions from the following: 2×10=20

(a) Give the geometrical interpretation of differential equation $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$, where P, Q, R are functions of x, y, z.

(b) Verify whether the differential equation $(yz + y^2)\,dx + (xz + z^2)\,dy + (y^2 - xy)\,dz = 0$ is integrable or not.

(c) Find the singular solution of the differential equation $p^2 + y^2 - 1 = 0$, where $p \equiv \frac{dy}{dx}$.

(d) Solve: $[D^2 - (a + b)D + ab]\,y = 0$, where $D \equiv \frac{d}{dx}$.

(e) Find the particular integral (P. I.) of the differential equation $[D^2 - a^2]\,y = \sin ax$.

(f) Solve the equation: $\frac{dx}{y} = \frac{dy}{x} = \frac{dz}{z}$

(g) State Picard's existence theorem for solving the differential equation: $\frac{dy}{dx} = f(x, y)$ with $y(x_0) = y_0$.

(h) Define singular solution of a differential equation.

(i) Show that the differential equation $M(x, y)dx + N(x, y)dy = 0$, possesses an infinite number of integrating factors.

(j) When an equation of the form $M(x, y)dx + N(x, y)dy = 0$ is said to be an exact differential equation?

(k) Solve the differential equation: $p^3 - 6p^2 + 11p - 6 = 0, p \equiv \frac{dy}{dx}$.

(l) Reduce the differential equation $(y + xp)^2 = x^2p$ to Clairaut's form by the substitution $xy = v$.

(m) Solve the differential equation: $\frac{d^2y}{dx^2} = xe^x$

(n) Find the differential equation of the circles passing through the origin and having their centres on x-axis.

(o) Give the geometrical interpretation of the differential equation $Pdx + Qdy + Rdz = 0$.

2. Answer *any four* questions: 5×4=20

(a) Find $f(y)$, if $z(1+y)dx - xzdy - xf(y)dz = 0$ is integrable. Find the corresponding integral. 3+2

(b) Using the method of variation of parameter, solve the differential equation

$y'' + y = \operatorname{cosec} x$.

(c) Solve: $x\frac{d^2y}{dx^2} - (2x-1)\frac{dy}{dx} + (x-1)y = 0$, given that $y = e^x$ is a solution.

(d) Show that the general solution of the differential equation $\frac{dy}{dx} + P(x)y = Q(x)$ can be written in the form $y = k(u-v) + v$, where k is a constant and u, v are two particular solutions of the equation.

(e) Solve the following system of differential equations:

$\frac{dx}{dt} = 4x - 2y, \frac{dy}{dt} = x + y$

(f) Solve the differential equation: $\frac{dx}{x+y} = \frac{dy}{x+y} = \frac{dz}{-(x+y+2z)}$

3. Answer *any two* questions: 10×2=20

(a) (i) Find a necessary condition of integrability of $Pdx + Qdy + Rdz = 0$, where P, Q, R are functions of x, y, z.

(ii) Solve: $\frac{dx}{z(x+y)} = \frac{dy}{z(x-y)} = \frac{dz}{x^2+y^2}$ 5+5

(b) (i) Solve: $[D^2 + a^2]\, y = \sin ax$

(ii) If u, v are two solutions of the differential equation $\frac{dy}{dx} + P(x)y = Q(x)$ and if $v = uz$, then show that $z = 1 + C.e^{-\int \frac{Q}{u} dx}$, where C is an arbitrary constant. 5+5

(c) (i) Solve: $(1+y^2)dx - (\tan^{-1} y - x)\, dy = 0$

(ii) Reduce the differential equation $x^2p^2 + y(2x+y)p + y^2 = 0$ to Clairaut's form by the substitution $y = u, xy = v$ and then solve it. 5+(3+2)

(d) (i) Solve: $(x^3D^3 - x^2D^2 + 2xD - 2)y = x^3$.

(ii) Show that the condition of integrability is satisfied for the equation

$(ydx + xdy)(a - z) + xydz = 0$. Hence solve it. 5+(2+3)